



Federated semi-supervised learning via globally guided pseudo-labeling: A robust approach for label-scarce scenarios

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ABSTRACT

Federated Semi-Supervised Learning (FSSL) is a powerful paradigm for collaboratively training models on both labeled and unlabeled datasets, which is adopted in domains such as healthcare and IoT. However, heterogeneous data distributions and imbalanced labeling capabilities both lead to significant prediction bias across participating clients, further resulting in skewed pseudo-labels during the local training stage. Most existing FSSL studies address the bias by improving model consistency, which relies on a well-trained benchmark derived from the fully labeled client, and encounters challenges in label-scarce scenarios. In this paper, we propose a novel FSSL method, namely Federated Globally Guided pseudo-labeling (FedGGp), suitable for both label-scarce and Non-Independent and Identically Distributed (Non-IID) scenarios. Specifically, this strategy summarizes the prediction bias assessments based on skewed class predictions, and modifies pseudo-labeling indicators accordingly in the subsequent iteration. For advantageous classes, FedGGp employs adaptive thresholds to generate high-quality pseudo-labels, while for discriminated classes, it expands the number of pseudo-labels to ensure balanced model training. Moreover, soft consistency regularization is applied to broaden the boundary of pseudo-labels for some underrepresented classes, which are typically ambiguous during classifications. The experimental results on four different datasets demonstrate that FedGGp outperforms various state-of-the-art methods.

1. Introduction

The capacity of Federated Learning (FL) to facilitate distributed model training while maintaining data privacy has led to its extensive adoption in a range of domains, including medical imaging (Ng et al., 2021), financial analysis (Reddy et al., 2024), IoT (Cao et al., 2024) and industrial iterative control system (Peng et al., 2024; Tao et al., 2023, 2024). Despite the significant attention FL has received, much of the existing research has focused on supervised learning scenarios that rely on fully labeled datasets. However, the rapid expansion of practical applications has made annotation of large datasets increasingly challenging and impractical (Jiao et al., 2023). To address this issue, recent studies have shifted efforts towards federated semi-supervised learning (FSSL), which combines both labeled and unlabeled data to collaboratively train a robust global model (Li et al., 2024; Wu et al., 2024; Yang et al., 2021).

In the practice of FSSL, annotation capabilities vary significantly across clients, with most operating under limited resources. This label-scarce limitation constantly manifests as inherent annotation deficiencies, which are further compounded by data heterogeneity (Jeong et al., 2021; Shang et al., 2023; Zhang et al., 2024). Specifically, there is a prevalent class imbalance across clients, where certain classes are markedly underrepresented during model training. Furthermore, some

clients might possess entirely unlabeled datasets. As a result of lacking annotation guidance, local model training on these clients is susceptible to prediction bias. In this study, we focus on an FSSL setting where clients suffer from both annotation deficiency and data heterogeneity, with a significant proportion of clients having predominantly unlabeled data. Compared to previous FSSL settings, our consideration encompasses the ‘Labeled on Partial Clients’ scenario (Liang et al., 2022; Liu et al., 2021), and further extends to the practical assumption of fully labeled clients.

Previous FSSL research has employed pseudo-labeling techniques on each client to leverage scattered unlabeled data (Jin et al., 2023). However, in the presence of widespread annotation deficiencies, these methods struggle to maintain balanced predictions when generating pseudo-labels (Li et al., 2023; Wei & Huang, 2024). As illustrated in Fig. 1, an experimental example demonstrates the prediction biases from both local and global perspectives, highlighting the model’s inclination towards specific classes. Locally, this imbalanced prediction leads to a bias towards certain classes during pseudo-label generation, which subsequently misguides classification in model training. More concerning, as the number of iterations increases, some classes are entirely excluded from pseudo-label predictions, creating a feedback loop that exacerbates prediction bias. A parallel issue arises in the aggregated global model,

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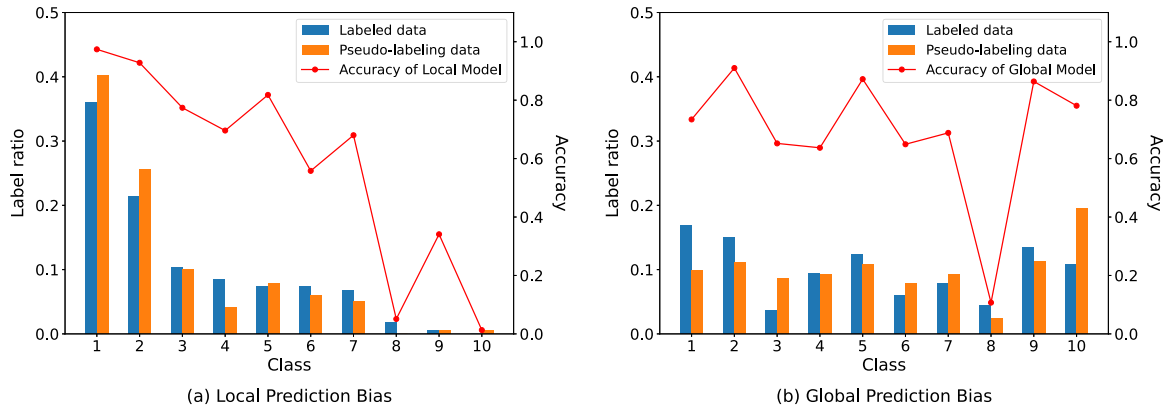


Fig. 1. Local and global prediction bias: The experiment integrates FedAvg with FixMatch on CIFAR10 dataset, with a labeled rate of 0.1, involving 10 clients and using a Dirichlet 0.3 to simulate data heterogeneity. The divergence distribution between labeled data and pseudo-labels, along with imbalanced prediction accuracies across different classes, demonstrates significant prediction bias.

where a comparable decline in performance across certain classes persists in classification tasks. Nevertheless, with the broader global view, this bias is somewhat moderated compared to the extreme skew observed in local predictions.

In response to the prediction bias in FSSL, researchers have explored various techniques, such as robust aggregation and gradient separation, to enhance model consistency (Liang et al., 2022; Zhang et al., 2024). While these consistency-based methods may offer some mitigation, they fall short of providing a comprehensive solution, especially for restricted FSSL cases. These methods predominantly depend on an effective consistency benchmark derived from fully labeled clients, which are often limited in practical settings. An approach that emphasizes fairness (Wei & Huang, 2024) employs class-balanced task allocation to ensure uniform training sets, yet its local pseudo-labeling remains susceptible to prediction bias. A recent study (Li et al., 2023) underscores that prior research has rarely addressed prediction bias directly in the context of pseudo-labeling, instead relying on the distribution of training samples to guide label assignment. However, in highly heterogeneous scenarios, this reliance on the imbalanced labeled distribution can exacerbate prior biases, ultimately resulting in performance degradation.

In this study, we propose Federated Globally Guided pseudo-labeling (FedGGp), which generates balanced pseudo-labels across classes even under both data heterogeneity and annotation deficiency. Concretely, FedGGp decouples local pseudo-labeling designations from biased local predictions by incorporating a global probability assessment. Compared to previous methods (Li et al., 2023; Liang et al., 2022) that rely on category distribution based guidance, bayesian calibration (Carlin & Louis, 2008) and exponential smoothing (Barrow et al., 2020) are employed to direct guidance within the prediction probability space. This approach mitigate prior bias and address the numerical instability observed in data heterogeneous scenarios. By constructing globally calibrated prediction probabilities, clients can assess class-wise prediction bias and apply class-balancing compensation via differentiated adaptive thresholds and regularization strategies during local pseudo-labeling. Drawing on principles from classical SSL methods (Sohn et al., 2020; Wang et al., 2023), strict filtering is enforced for advantageous classes, while greater attention is paid to underrepresented classes, which frequently serve as a primary cause of performance degradation. Besides, an additional imbalance metric is introduced to further refine the pseudo-labeling strategy for those classes. Notably, for severely prediction biased classes, a soft consistency regularization mechanism is applied to recover low-confidence pseudo-labels near decision boundaries, preserving the balanced representation contributions in each training iteration. Extensive experiments conducted on four benchmark datasets across two FSSL scenarios validate the robustness and effectiveness of our method.

Overall, The main contributions of FedGGp are as follows:

- In this work, we propose a novel global probability assessment to estimate class prediction bias for guiding local pseudo-labeling. This assessment rebalances predicted probabilities with prior knowledge and derives independent class prediction tendencies.
- We propose a hierarchical pseudo-labeling strategy based on the prediction tendencies across classes, which synergistically integrates globally guided adaptive thresholds with a specifically designed soft consistency regularization mechanism, effectively expanding pseudo-labels of the underrepresented classes.
- Extensive experiments validate the superiority of FedGGp over state-of-the-art methods in the realm of FSSL.

2. Related work

2.1. Federated learning

Federated learning achieves collaborative model training across multiple clients while ensuring the privacy and security of local data. Among the key challenges in FL, data heterogeneity has garnered significant attention from the research community (Li et al., 2022, 2020b). FedAvg (McMahan et al., 2017), the most widely used baseline in FL research, facilitates collaborative model training by employing weight averaging during server aggregation to address data heterogeneity. This issue has been further explored in subsequent research, incorporating approaches such as initializing model consistency regularization to mitigate client drift (Acar et al., 2021; Li et al., 2020a), or introducing additional machine learning paradigms just like knowledge distillation Wu et al. (2022), multi-task learning (Mills et al., 2020; Smith et al., 2017), and meta-learning (Yang et al., 2023). Recently, research on prototypes and contrastive learning has demonstrated notable superiority, as it provides stable anchors in feature alignments (Dai et al., 2023; Tan et al., 2022). However, these methods rely on supervised training with labeled datasets, which renders them inapplicable when most data are unlabeled.

2.2. Semi-supervised learning

Pseudo-labeling is a prevalent SSL (semi-supervised learning) approach that generates predictive labels on unlabeled data, aiding in improving the generalization of model training (Arazo et al., 2020; Nguyen et al., 2023). FixMatch (Sohn et al., 2020) integrates fixed threshold pseudo-labeling with consistency regularization (Verma et al., 2019),

refining pseudo-labels through a dual-filtering mechanism. FreeMatch (Wang et al., 2023) points out that a fixed threshold exacerbates imbalanced class learning and presents adaptive thresholds for different classes, which achieves superior performance. Furthermore, other related studies have been conducted, including SoftMatch (Chen et al., 2023), which achieves a trade-off between the quality and quantity of pseudo-labels, and PRG (Duan et al., 2023), which proposes pseudo-label correction guidance for the heterogeneity of labeled and unlabeled data. Moreover, some of the pseudo-labeling studies (Jiang et al., 2024; Lin et al., 2024; Zhang et al., 2020) have been conducted from the perspective of fairness, with these methods placing greater emphasis on ensuring equitable class distribution in pseudo-labeling process. Although these methods perform well in SSL, directly integrating them into FSSL often yields suboptimal results. This is due to the fact that methods designed for centralized scenarios struggle to handle the locally extreme imbalances and heterogeneous data distributions inherent in federated settings.

2.3. Federated semi-supervised learning

Unlike SemiFL (semi-supervised federated learning) that associates labels with server (Lin et al., 2021; Long et al., 2021), FSSL is oriented towards scenarios where labels are stored on clients. In accordance with the stated objectives, FSSL is subdivided into two settings. (1) Labeled on Partial Clients: The majority of clients are unlabeled, while a few have fully labeled datasets, primarily concerned with the local training of unlabeled clients for collaborative learning (Li et al., 2023; Liang et al., 2022; Yang et al., 2021). (2) Clients with Partially Labeled: Each client has partially labeled data, with a primary focus on addressing the heterogeneity between labeled and unlabeled data within the client (Jeong et al., 2021; Zhu et al., 2024).

In this paper, we mainly focus on the former concern, where several works have been proposed. For instance, FedIRM (Liu et al., 2021) employs a relational matching module to build inter-client consistent predictions; RSCFed (Liang et al., 2022) computes aggregation weights of the sub-consensus based on model consistency; Twin-Sight (Zhang et al., 2024) decouples learning objectives into two models to avoid the gradient conflict between labeled and unlabeled data, which ensures consistency between supervised and unsupervised training. Despite extensive research conducted on maintaining consistency, these methods serve only to provide a palliative patch for biased pseudo-labels. The recent study, CBAFed (Li et al., 2023), elucidates the root issue of pseudo-label imbalance prediction and adapts thresholds based on disparate categories of pseudo-label ratios. Nevertheless, the selected pseudo-labels tend to diverge from the model's prediction bias to a certain extent, making this approach unsuitable for scenarios with high data heterogeneity. In parallel, FedFAPL (Wei & Huang, 2024) introduces a fairness-aware pseudo-labeling mechanism by assigning class-balanced tasks per round, but it is struggles to counteract initial prediction bias during early training stages.

In technical contrast to prior studies, FedGGp leverages the rebalanced prediction probability space rather than simply relying on sample frequency statistics for pseudo-labeling guidance. It constructs hierarchical strategies tailored to class-wise prediction skewness, introducing additional soft regularization techniques that are particularly effective when the model has low prediction confidence. Moreover, recent studies have examined privacy issues in FSSL, particularly regarding the use of globally shared pseudo-labels (Abourayya et al., 2025). FedGGp circumvents these concerns as it is independent of public dataset, and is readily compatible with standard FL privacy protections (Bonawitz et al., 2016; Wei et al., 2020).

3. Federated globally guided pseudo-labeling

In this section, we delineate the problem setting of FSSL and present our proposed method. FedGGp adheres to the conventional FedAvg

workflow, with the server maintaining a set of vectors to estimate prediction bias during training. For training with unlabeled data, FedGGp utilizes a meticulously designed pseudo-labeling strategy based on mapped global prediction evaluation, thereby guiding the local predictions through a global prediction probability space. Compared with previous methods, this design decouples local prediction biases from the pseudo-labeling strategy, mitigating the impact of skewed local predictions. The pseudo-labeling strategy integrates statistically adaptive thresholds and soft consistency regularization, which captures continuous encodings and constructs a hierarchical system to address the skewed class discrimination in FSSL. The overview of FedGGp is presented in Fig. 2.

3.1. Problem setup

We first define the FSSL scenario, which is grounded in data heterogeneity settings among clients. Consider K clients engaged in collaborative model training, where each client $k \in K$ holds a heterogeneous dataset D_k drawn from the different local distributions P_k . Due to varying annotation capacities, each client k has a labeled subset of data $D_k^l = \{x_k^i, y_k^i\}_{i=1}^{N_k^l}$, alongside a larger pool of unlabeled data $D_k^u = \{x_k^i\}_{i=1}^{N_k^u}$, such that $D_k = \{D_k^l, D_k^u\}$. This study primarily focuses on label scarcity in highly heterogeneous data, where unlabeled clients widely exist, and fully labeled clients are either skewed or excluded.

In each iteration, the client receives model parameters broadcast by the server, denoted as θ , and proceeds to train the model using its local dataset. For the labeled data, classical supervised learning is employed to facilitate categorical classification. For unlabeled data, pseudo-labeling is applied to augmented samples, generating predicted labels with the maximum probability, denoted as $\hat{y}$:

$$\hat{y} = \arg \max p_\theta(y|x_{\text{aug}}) \quad (1)$$

where x_{aug} is an augmented sample of x , with aug referring to common data augmentation strategies (Shorten & Khoshgoftaar, 2019).

Accordingly, the predicted probability corresponding to $\hat{y}$ can be denoted as $\hat{p} = \max p_\theta(y|x_{\text{aug}})$. By employing the threshold selection strategy, high-confidence pseudo-labels suitable for model training can be effectively derived. For fixed-threshold selection, the local optimization objective of FSSL can be expressed through the empirical risk:

$$\min_{\theta} \mathcal{L} = \underbrace{\frac{1}{|D_k^l|} \sum_{(x,y) \in D_k^l} L_s((x,y), \theta)}_{\text{labeled}} + \lambda \underbrace{\frac{1}{|D_k^u|} \sum_{x \in D_k^u} \mathbb{I}(\hat{p} > \tau) \cdot L_u((x, \hat{y}), \theta)}_{\text{unlabeled}} \quad (2)$$

L_s and L_u are the loss functions for labeled and unlabeled training, typically utilizing cross-entropy. τ provides the fixed selection threshold and λ is the balance coefficient between the two loss terms.

3.2. Global prediction assessment

As previously discussed, a fundamental limitation of SSL pseudo-labeling methods is the inability to adapt to the skewed view of federated clients. Local training with imbalanced classes results in the acquisition of distorted knowledge, which subsequently impairs biased model predictions. To address this issue, we propose an evaluation of prediction bias from a global perspective, which offers a more balanced view compared to local assessments.

Related research indicates that the effectiveness of classification can be assessed by the prediction confidence of the calibrated model (Guo et al., 2017; Wang et al., 2023). To evaluate the bias of model predictions across classes, a global prediction evaluation vector, denoted as Φ , is established in the server. The definition of Φ is as follows:

$$\Phi(c) = \frac{1}{|D^u|} \sum_{x \in D^u} p_c(x) \quad (3)$$

$c \in C$ represents a class within the class set, D^u denotes the unlabeled dataset, and $p_c(x)$ indicates the confidence that sample x belongs to class

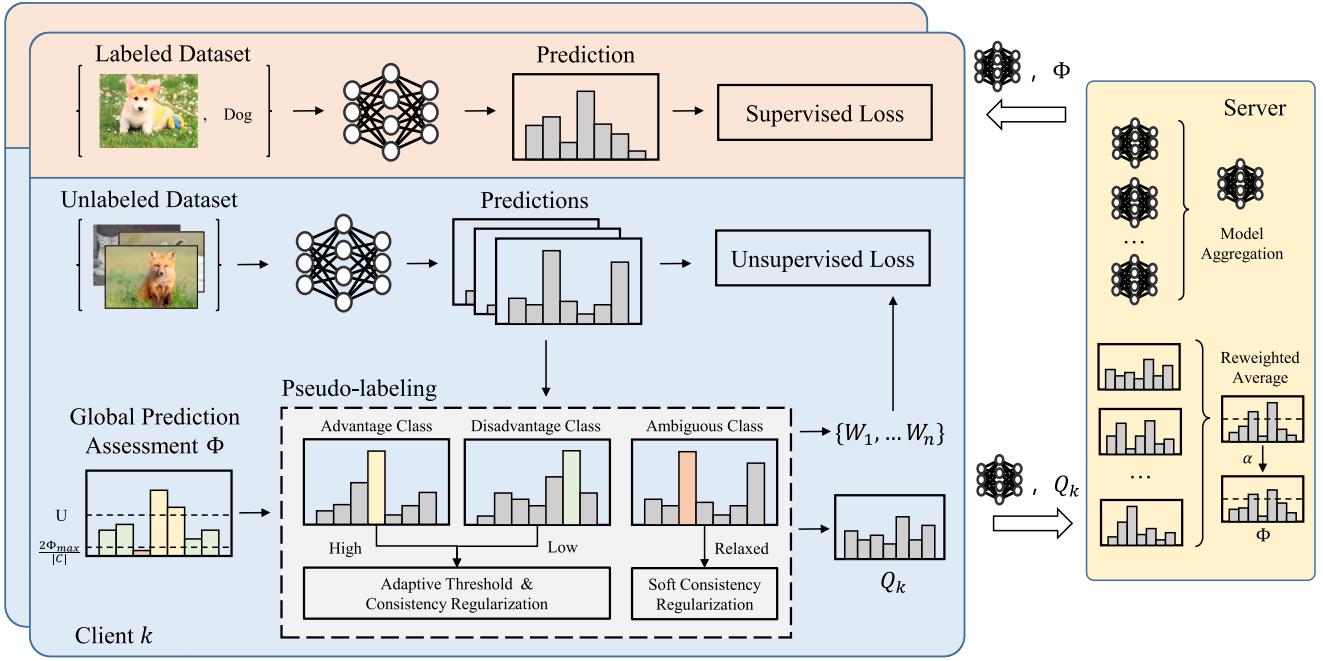


Fig. 2. The overview of FedGGp. The blue background section illustrates the globally guided pseudo-labeling strategy, while the pseudo-labeling process is outlined within the dashed box. Here, Φ denotes the global prediction evaluation vector, and C represents the set of classification classes. The uniform distribution U and $\frac{2\Phi_{\max}}{|C|}$ respectively define the boundaries for class partitioning. The primary pseudo-labeling output is $\{W_1, \dots, W_n\}$, which serves as the weighting coefficients assigned to pseudo-labels for n unlabeled data points. Q_k represents the accumulated prediction probability for each sample obtained through local pseudo-labeling, specifically designed to minimize communication costs. The symbol α is a specially designed rebalancing factor used for Q_k aggregation in the server.

c . Given the substantial communication resources required by p , in practice, clients typically compute and transmit the accumulated prediction vector $Q = \sum_{x \in D^u} p(x)$ as an alternative.

In Eq. (3), the model yields fair predictions for each class when the samples in D^u are balanced across classes, in which case the global evaluation vector $\Phi(c)$ approximates the uniform distribution. However, due to imbalanced distributions and class prediction bias, the empirical distribution of Φ exhibits significant discrepancies. To mitigate this, we refine the evaluation process by replacing the full unlabeled set D^u with the filtered pseudo-labeled set D^p , which contains samples with higher prediction confidence and is more consistent with reliable classification behavior. Nevertheless, this stricter selection amplifies the skew in the prior distribution, since underrepresented classes typically fail to yield sufficiently confident predictions. According to Bayesian theory (Bernardo & Smith, 2009; Blei et al., 2017; Carlin & Louis, 2008), such prior bias can distort posterior confidence estimation. Thus, we introduce a rebalancing factor α to calibrate class-wise accumulation confidence by compensating for prior imbalance induced by D^p . Formally:

$$\Phi(c) = \frac{1}{\alpha} \cdot \frac{1}{|D^p|} \sum_{x \in D^p} p_c(x) \quad (4)$$

Defining $P(U)$ as a uniform reference, we explore three design schemes for the rebalancing factor α :

- (1) $\alpha_1 = P(c)/P(U)$, which strictly follows Bayesian by adjusting the posterior with empirical class prior.
- (2) $\alpha_2 = \exp(P(c))$, which introduces basic exponential smoothing to empirical class prior.
- (3) $\alpha_3 = \exp(P(C) - P(H))$, which applies relative deviation exponential scalings based on empirical class prior and uniform reference.

Among them, α_1 provides a theoretical solution based on classic Bayesian theory. However, in FSSL scenarios with severe prediction imbalance, α_1 suffers from catastrophic numerical instability or zero-value edge effects. To address these issues, α_2 and α_3 serve as smoothed extensions grounded in Bayesian calibration theory (Vaicenavicius et al.,

2019). The distinction between those two lies in that α_3 introduces a discriminative correction term, enabling both downscaling and upscaling. The effectiveness of each design is empirically compared in the experimental section, where α_3 is adopted as the default due to its superior performance under extreme heterogeneity.

A practical recommendation is to utilize $P(U)$ to instantiate Φ and update with high momentum β (typically set to 0.9 or 0.99), which maintains the optimization consistency in model training. Combining with Eq. (4), the update process of Φ in round t can be represented as follows:

$$\Phi^t(c) = \beta \cdot \Phi^{t-1}(c) + (1 - \beta) \cdot \Phi(c) \quad (5)$$

3.3. Globally guided pseudo-labeling strategy

Classical pseudo-labeling strategies (Sohn et al., 2020; Wang et al., 2023) have demonstrated robust performance by incorporating thresholds (T.) and consistency regularization (C.R.) through a dual screening selection process. Specifically, C.R. (Laine & Aila, 2017) enforces prediction consistency across different perturbations of the same input, strictly filtering out samples with unstable predictions. This dual pseudo-labeling mechanism can be expressed as follows:

$$W(x) = \underbrace{\mathbb{1}(\hat{p}^1 > \tau)}_{T.} \cdot \underbrace{\mathbb{1}(\hat{y}_1 = \hat{y}_2)}_{C.R.} \quad (6)$$

In this equation, $\hat{y}_1$ and $\hat{y}_2$ denote the predicted classes with the highest confidence obtained from two perturbed views of the same input, and $\hat{p}^1$ is the corresponding confidence of $\hat{y}_1$. The function $W(x)$ mathematically serves as an indicator function determining whether a sample's prediction should be utilized as a pseudo-label. However, skewed data training frequently results in a severe underrepresentation of specific tail classes, complicating the application of those approaches in FSSL scenarios. To this end, we introduce a tailored pseudo-labeling scheme that incorporates globally guided adaptive thresholds and soft consistency regularization.

Globally guided adaptive threshold. To better adapt pseudo-labeling strategies to the skew across different classes, we introduce the Imbalance Factor (IF) from long-tailed studies (Cao et al., 2019), and extend it to the realm of pseudo-label probabilities. Specifically, we apply a uniform baseline and an IF boundary to $\Phi(c)$ to quantify the degree of prediction bias across classes. Based on this guidance, the class set C can be dynamically partitioned into three distinct groups, each of which is defined as follows:

- **Advantageous classes (A):** Classes with prediction probabilities exceeding the uniform prior baseline, typically receiving the majority of pseudo-labels due to high-confidence outputs.
- **Disadvantageous classes (D):** Classes with biased prediction probabilities within the IF imbalance boundary, allowing partial sample inclusion in pseudo-labeling.
- **Ambiguous classes (Am):** Classes with severely suppressed prediction probabilities below the IF imbalance boundary, resulting in most samples being excluded from pseudo-labeling.

Here, we set the default boundary $IF = \frac{|C|}{2}$ to adaptively distinguish different datasets with varying degrees of imbalance. While reminiscent of head-tail partitioning, this categorization is based on pseudo-label prediction probabilities rather than sample frequencies. Formally, defining $\Phi_{\max} = \max_{c \in C} \Phi(c)$, the specific category stratification can be described as follows:

$$C = \begin{cases} \mathbf{A} & \text{if } \Phi(c) \geq \frac{1}{|C|}, \\ \mathbf{D} & \text{if } \frac{1}{|C|} > \Phi(c) \geq \frac{2\Phi_{\max}}{|C|}, \\ \mathbf{Am} & \text{if } \Phi(c) < \frac{2\Phi_{\max}}{|C|}. \end{cases} \quad (7)$$

We establish basic thresholds based on the predictions from the mini-batch and adjust thresholds with global prediction evaluation. Specifically, for $c \in \mathbf{A}$, the corresponding adaptive threshold is defined as $\tau_A = \frac{\Phi(c)}{\Phi_{\max}} \cdot \mathbb{E}_B(\hat{p})$, where $\mathbb{E}_B(\hat{p})$ represents the expected maximum prediction confidence $\hat{p}$ across all samples in the corresponding batch. For $c \in \mathbf{D}$, the calibrated confidence estimates remains below standard predicted probabilities. To capture more of the pseudo-label predictions, we further reduce the adaptive threshold, defined as $\tau_D = \frac{\Phi(c)}{\Phi_{\max}} (\mathbb{E}_B(\hat{p}) - \sigma_B(\hat{p}))$, where $\sigma_B(\hat{p})$ represents the standard deviation of the set of confidence scores $\hat{p}$. The significant challenge for the ambiguous class $\mathbf{Am}$ is the severe classification confusion among samples. This misclassification makes it difficult to discern via both threshold selection and consistency regularization. Therefore, we establish the lowest classification threshold for $\mathbf{Am}$, defined as $\tau_{Am} = \frac{1}{|C|}$, and further explore pseudo-label predictions from the perspective of consistency regularization.

Note 3.1: If the range of $\mathbb{E}_B(\hat{p})$ falls within $(\frac{1}{1+\sqrt{3}}, 1)$, then the inequality $\tau_A > \tau_D > \tau_{Am}$ holds, and the threshold setting for category c is positively correlated with $\Phi(c)$.

Proof. Given that $\sigma_B(\hat{p}) > 0$, inequality $\tau_A > \tau_D$ can be directly derived from:

$$\mathbb{E}_B(\hat{p}) > \mathbb{E}_B(\hat{p}) - \sigma_B(\hat{p})$$

For $\tau_D > \tau_{Am}$, based on the partition, the inequality can simplify to:

$$\frac{\Phi(c)}{\Phi_{\max}} (\mathbb{E}_B(\hat{p}) - \sigma_B(\hat{p})) > \frac{1}{|C|}$$

Substituting the boundary into the inequality and simplifying yields, we have $\mathbb{E}_B(\hat{p}) > \frac{1}{2} + \sigma_B(\hat{p})$. In the worst-case analysis, assume that $\hat{p}$ is uniformly distributed within the interval $(l, 1)$. Given by the expectation and standard deviation of $\hat{p}$, we have:

$$\frac{l+1}{2} > \frac{1}{2} + \frac{1-l}{\sqrt{12}}$$

Simplifying the inequality yields the lower bounds for l and $\mathbb{E}_B(\hat{p})$, and then the inequality $\tau_A > \tau_D > \tau_{Am}$ is confirmed. $\square$

A detailed derivation can be found in the appendix. In practice, $\mathbb{E}_B(\hat{p})$ is typically close to 1 and $\mathbb{E}_B(\hat{p}) \gg \sigma_B(\hat{p})$. To prevent potential anomalies, a lower boundary is configured, expressed as $\mathbb{E}_B(\hat{p}) = \max(\mathbb{E}_B(\hat{p}), 0.5)$.

Soft consistency regularization. The traditional consistency regularization methods (Laine & Aila, 2017) represent sample weights as one-hot encoded vectors, such as $\mathbb{1}_{\hat{y}_1 = \hat{y}_2}$. This technique effectively excludes the majority of ambiguous samples. However, in the presence of prediction bias, consistency regularization exacerbates unfairness towards currently underrepresented classes (e.g. class $\mathbf{Am}$), as it tends to disregard most samples within these classes.

To further leverage these previously discarded samples for balanced model training, we propose a soft consistency regularization method to recover underrepresented class samples, as shown in Fig. 3. This approach extends the one-hot encoded vector to a continuous form, assigning weights based on classification consistency confidence, formulated as $R_s := \frac{p_{c \leftarrow \hat{y}_1}^2}{p_{c \leftarrow \hat{y}_2}^2}$. Here, $p_{c \leftarrow \hat{y}_1}^2$ denotes the second prediction probability for class $\hat{y}_1$. Notably, when $\hat{y}_1 = \hat{y}_2$, the encoded indicator of soft consistency regularization aligns with that of traditional consistency regularization.

However, unrestricted sample recovery may lead to an increase in low-quality or misclassified pseudo-labels. Thus, we introduce a Top-k mechanism as a controllable secondary constraint to filter out predictions with lower confidence. The continuous encoded vector for soft consistency regularization is given by:

$$R_s(\hat{y}_1) := \frac{p_{c \leftarrow \hat{y}_1}^2}{p_{c \leftarrow \hat{y}_2}^2} \cdot \mathbb{1}_{\text{Top-k}(p_{c \leftarrow \hat{y}_2}^2, \kappa)} \quad (8)$$

where κ serves as the hyperparameter of Top-k. Furthermore, we retain the use of traditional consistency regularization methods for $\mathbf{A}$ and $\mathbf{D}$, represented as $R_h(\hat{y}_1) := \mathbb{1}(\hat{y}_1 = \hat{y}_2)$.

In summary, the process for predicting pseudo-label weights is expressed as follows:

$$W_{ggp}(x, \Phi) = \begin{cases} \mathbb{1}(\hat{p}_1 > \tau_A) \cdot R_h(\hat{y}_1), & \hat{y}_1 \in \mathbf{A}, \\ \mathbb{1}(\hat{p}_1 > \tau_D) \cdot R_h(\hat{y}_1), & \hat{y}_1 \in \mathbf{D}, \\ \mathbb{1}(\hat{p}_1 > \tau_{Am}) \cdot R_s(\hat{y}_1), & \hat{y}_1 \in \mathbf{Am}, \end{cases} \quad (9)$$

3.4. Pipelines of FedGGp

The final training objective for client k is to minimize the following function:

$$\mathcal{L} = \frac{1}{|D_k^a|} \sum_{(x,y) \in D_k^a} L_s((x, y), \theta) + \lambda \frac{1}{|D_k^u|} \sum_{x \in D_k^u} W_{ggp}(x, \Phi) \cdot L_u((x, \hat{y}), \theta) \quad (10)$$

Algorithm 1 provides a detailed algorithmic process of FedGGp. Based on these improvements, we further elaborate on the framework's broader implications, particularly in terms of communication efficiency, privacy guarantees, and potential application scenarios.

Communication. Communication costs associated with FedGGp are primarily attributable to the transmission of category prediction confidence. However, transmitting the prediction confidence of all samples during training is impractical. Instead, the cumulative mean confidence is calculated on the client side, denoted as Q_k^t , and re-weighting is performed on the server side. The length of Q_k^t corresponds to the number of categories, and its communication costs associated with the transmission of the model are negligible.

Privacy and robustness. FedGGp follows the standard federated framework and maintains compatibility with existing privacy-preserving techniques, such as secure aggregation (Bonawitz et al., 2016) and differential privacy (Wei et al., 2020). Moreover, the transmission of prediction vectors provides inherent privacy advantages similar to prototype-based methods in Non-IID research (Tan et al., 2022), while containing less sensitive information. For robustness, FedGGp is inherently designed as a robust solution for label-deficient scenarios and

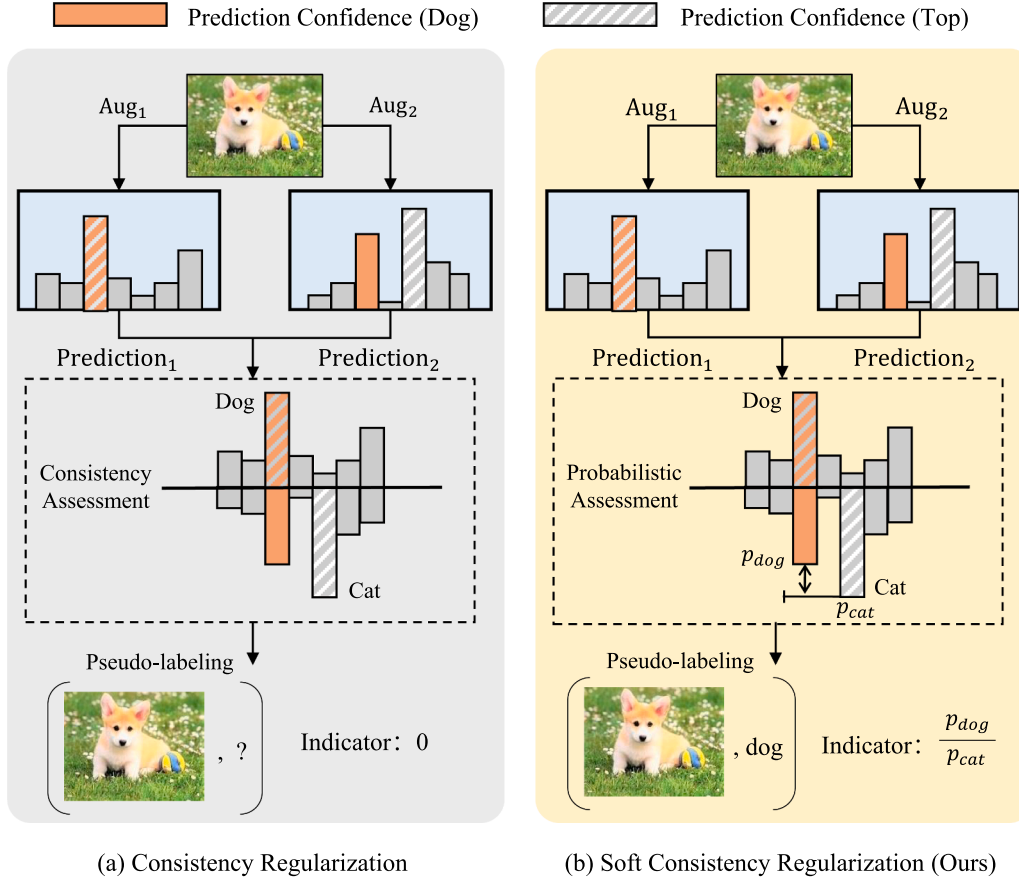


Fig. 3. A comparison of soft consistency regularization and traditional consistency regularization for pseudo-labeling. In this example, samples of the “dog” are difficult to distinguish and may be misclassified as belonging to the “cat”. The Symbol p_{dog} and p_{cat} denote the predicted probabilities that the current sample belongs to dog and cat classes, respectively.

its architecture natively supports integration of client validation mechanisms (Bao et al., 2024; So et al., 2020) to address practical robustness challenges, including noisy or Byzantine clients when present in real-world deployments.

Potential application. While FedGGP is validated on image classifications, its design is modality-agnostic and naturally extends to broader federated semi-supervised scenarios. Beyond vision, FedGGP is well-suited for practical applications such as medical time-series classification (Ju et al., 2020), personalized recommender systems (Yang et al., 2019), and industrial IoT anomaly detection (Meyer et al., 2023). In these diverse domains, labeling rare events is prohibitively costly, and the data distribution is highly skewed and sparse, which makes FedGGP particularly suitable.

4. Experiments

4.1. Experimental setup

Datasets and models. Building on previous research (Li et al., 2023), we evaluate FedGGP on four image classification datasets: CIFAR10 (Krizhevsky & Hinton, 2009), CIFAR100 (Krizhevsky & Hinton, 2009), SVHN (Netzer et al., 2011), and Fashion MNIST (Xiao et al., 2017). For all datasets, a lightweight ResNet network is employed as model backbone (He et al., 2016; Shang et al., 2023). This network has fewer parameters than ResNet18 and is more suitable for smaller datasets. The same model architecture is implemented across all compared FL methods to ensure a fair comparison.

FSSL settings. In relation to the classical data heterogeneity setting (Hsu et al., 2019), we simulate high data heterogeneity using a Dirichlet distribution, denoted as $Dir(\delta)$. Here, δ is set to 0.3, a typical configuration for high heterogeneity studies. For client labeling, we define a triplet (x, y, z) , where x , y , and z respectively represent the number of fully labeled clients, partially labeled clients, and unlabeled clients. It is assumed that the labeled dataset is class-imbalanced both globally and locally, as all labeled data is randomly sampled from the clients. The labeling rate for partially labeled clients is set to 0.05.

Following (Li et al., 2023), we configure 10 clients, all of which participate in each communication round. Specifically, we apply two different (x, y, z) configurations: (1) The triplet (1, 2, 7) is similar to the commonly used setup with one fully-labeled client, where the inclusion of two partially labeled clients ensures that most classes are covered by labeled data. (2) The triplet (0, 5, 5) serves as the default configuration for subsequent studies, where the absence of fully-labeled clients and the presence of a limited number of labels on some clients more closely reflect real-world conditions. Both configurations are illustrated in the Fig. 4.

Baselines and competitors. We compare FedGGP against the following methods, mainly including a standard baseline and two categories of competitors:

- **Standard Baseline:** A basic lower bound, referred to as FedAvg-Lower (McMahan et al., 2017), is established by applying FedAvg with supervised training on labeled datasets.
- **FedAvg with SSL Methods:** We incorporate recent SSL methods combined with local training via FedAvg, including FedAvg + FixMatch

Algorithm 1: Pipelines of FedGGp.

Input: number of clients K ; number of classes C ; local epochs E ; learning rate η ; communication rounds T ; datasets D^a and D^u .

Output: global model θ^T .

```

1 Server Executes:
2 for  $t = 1$  to  $T$  do
3   initialize model parameters as  $\theta^0$  and set  $\Phi^0$  as uniform
4   Broadcast  $\theta^{t-1}$  and  $\Phi^{t-1}$  to clients
5   for each active client  $k \in K$  do
6      $\theta_k^t, Q_k^t \leftarrow \text{LocalTraining}(k, \theta^{t-1}, \Phi^{t-1})$ 
7   end
8   Aggregate  $\theta^t = \sum_{k \in \mathcal{M}} \frac{|D_k|}{|D|} \theta_k^t$ 
9   Update  $\Phi^t$  with  $Q_k^t$  from Eq.(3) and Eq.(4)
10 end
11 return  $\theta^T$ 
12
13 LocalTraining( $k, \theta^{t-1}, \Phi^{t-1}$ )
14 begin
15   Initialize  $\theta_k^t \leftarrow \theta^{t-1}$ 
16   Initialize  $Q_k^t$  as an empty vector of length  $C$ 
17   Classify categories  $A$ ,  $D$  and  $A_m$  based on  $\Phi^{t-1}$  using Eq.(7)
18   for local epoch  $e = 1$  to  $E$  do
19     for each mini-batch  $B^u$  in  $D_k^u$  and  $B^a$  in  $D_k^a$  do
20       Perform supervised training on  $B^a$ 
21       Collect outputs  $p^1$  and  $p^2$  on  $B^u$ 
22       Predict pseudo-labels with  $W_{ggp}(x_{B^u}, \Phi^{t-1})$ 
23        $w_k^t \leftarrow w_k^t - \eta \nabla_w \mathcal{L}(B^a, B^u)$ , where  $\mathcal{L}$  is defined in Eq.(10)
24     end
25     Construct pseudo-label set  $D^p$  based on  $W_{ggp}$ 
26     Collect  $p^1$  from samples in  $D^p$  and stack into  $Q_k^t$ 
27   end
28   return  $\theta_k^t, Q_k^t$ 
29 end

```

(Sohn et al., 2020), FedAvg + FreeMatch (Wang et al., 2023), and FedAvg + SoftMatch (Chen et al., 2023).

- Latest FSSL Methods: The latest FSSL methods such as RSCFed (Liang et al., 2022), CBAFed (Li et al., 2023), and Twin-Sight (Zhang et al., 2024), which are compatible with training on both labeled and unlabeled clients.

Implementation details. For the implementation of those algorithms, we use stochastic gradient descent (SGD) as the default optimizer, a mini-batch size of 64, the learning rate is fixed at 0.003, momentum 0.9, and weight decays $5e - 4$. The number of local training epochs is 1 for all experiments. We utilize the hyperparameters specified in the original paper for the above baselines. For hyperparameters of FedGGp, the value of κ in Top-k is set to 2 for CIFAR10 and SVHN, while 5 for CIFAR100. Following the previous study (Zhang et al., 2024), we set different loss coefficients, λ , corresponding to unlabeled and partially labeled clients: values of 0.1 and 3 for the (1, 2, 7) scenario, 0.5 and 3 for the (0, 5, 5) scenario. For all experiments, we set a total communication round limit of 500 and recorded the optimal model during the training process.

4.2. Comparisons with state-of-the-arts methods

Method evaluation. From the experimental results in Table 1, FedGGp consistently ranks among the top methods compared to other state-of-the-art approaches and, in most cases, demonstrates superior

performance. Its advantages are more evident in complex tasks like CIFAR10 and CIFAR100, as opposed to simpler ones like Fashion-MNIST and SVHN. This refers to its refined pseudo-labeling strategy and data recovery mechanism for discarded samples. However, these benefits come with the trade-off of longer convergence rounds for complex tasks, as shown in Fig. 5. Furthermore, FedGGp displays a higher degree of convergence variability. It is worth noting that, with the exception of CBAFed, which utilizes a fixed pseudo-labeling strategy, adaptive thresholding methods such as FreeMatch and Twin-Sight also exhibit comparable convergence variability. The method with the highest fluctuations, RSCFed, is affected by client sampling rather than adaptive thresholds.

For other methods, since labeled and unlabeled data follow the same distribution, SSL methods sometimes demonstrate acceptable performance. However, those approaches exhibit significant variation across different datasets and scenarios. The FSSL method RSCFed, which relies on sampler and consensus, has been observed to experience difficulties in adapting to extreme label scarcity. CBAFed has been identified to depend heavily on the warm-up model with fully labeled clients, which impedes maintaining performance under high heterogeneity. Twin-Sight exhibits excellent overall performance, with only suboptimal results on Fashion MNIST, indicating that the gradient discrepancy between supervised and unsupervised training has a diminished impact on simpler tasks.

Heterogeneity comparison. We conduct further experiments to investigate the impact of data heterogeneity by adjusting $\delta \in \{0.1, 0.3, 0.8\}$ on CIFAR10. Testing methods include FedAvg-lower bound as a baseline, and well-performing methods Twin-Sight, FedAvg + FixMatch, and FedAvg + SoftMatch as comparisons.

As illustrated in Table 2, there is a discernible trend of diminished performance across all methods as data heterogeneity increases. While FedGGp demonstrates a distinct advantage in scenarios with high data heterogeneity, it exhibits suboptimal performance in IID settings. The performance degradation observed in FedGGp fundamentally originates from its architectural emphasis on mitigating the negative impact of local prediction bias on local pseudo-labeling. This design paradigm, while essential for correcting skewed distributions in Non-IID, proves inconsequential under IID conditions where local prediction bias becomes negligible during local training. This performance gap is further compounded by inherent algorithmic preference in FedGGp for expanding under-represented class samples, whereas IID conditions naturally emphasize sample quality over quantity. Notably, the performance degradation in IID settings is consistently observed in SoftMatch, which similarly employs pseudo-labeling scope expansion, providing strong empirical validation for our interpretation. Moreover, through systematic hyperparameter optimization, FedGGp demonstrates adaptability by achieving 84.00% accuracy with increased β and disabled soft consistency regularization, just suboptimal to Twin-Sight's 84.15% benchmark. While FedGGp shows marginally suboptimal performance in IID settings, it demonstrates clearly superior performance in Non-IID scenarios. These results provide compelling evidence for FedGGp's robust generalization capabilities across diverse data distributions.

Computational and communication overhead. Table 3 systematically compares the system overhead across those evaluated methods, including (1) per-client communication cost, (2) computational complexity (FLOPs), and (3) execution time. Both absolute measurements and relative ratios are provided for comprehensive comparison.

Notably, FSSL methods consistently demonstrate significantly higher computational complexity compared to the fully-supervised FL baseline (denoted as FedAvg-labeled), with this overhead primarily attributable to consistency regularization. In contrast to conventional FSSL approaches, FedGGp achieves competitive efficiency in both computational and communication aspects. Specifically, FedGGp introduces less than 0.01% additional overhead for probability threshold computation and communication, since model operations constitute the predominant system overhead. This efficiency advantage becomes particularly

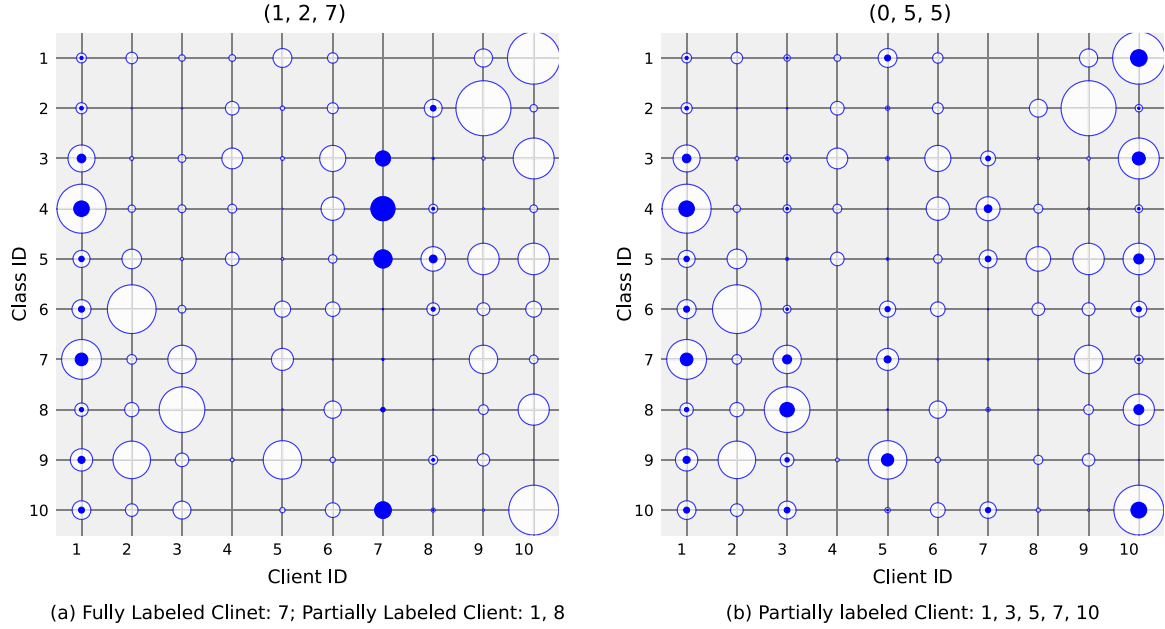
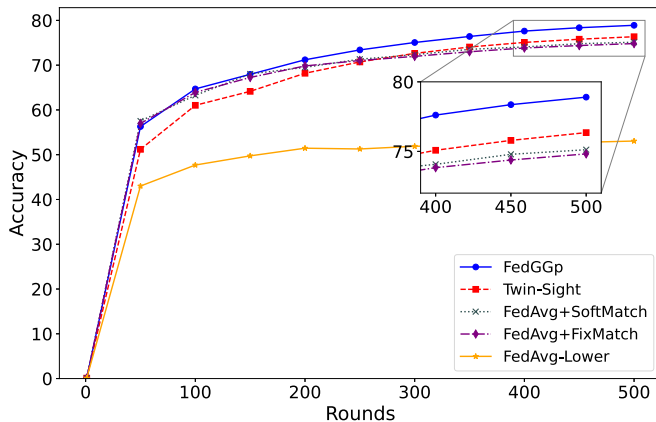


Fig. 4. Visualization of data distributions of CIFAR10 with the two different FSSL settings. “○” dots represents unlabeled data, while “●” dots means labeled data. The size of each dot reflects the number of samples in each client. (1,2,7) indicates 1 fully labeled, 2 partially labeled, and 7 unlabeled clients; (0,5,5) indicates 5 partially labeled and 5 unlabeled clients.

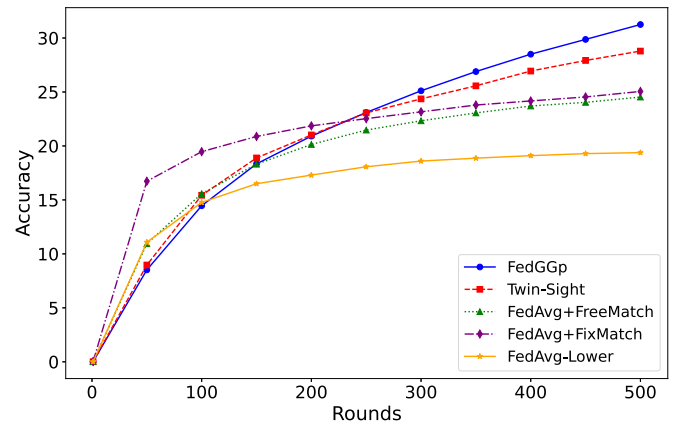
Table 1

Results on the Fashion MNIST, CIFAR10/100, and SVHN datasets under heterogeneous data conditions, recording both the accuracy and convergence fluctuations of the compared methods. The Best method in each setting is highlighted in bold fonts, and the second place is underlined for prominence. The symbol * indicates that some methods extend training rounds for further model convergence.

Method	Fashion MNIST		CIFAR10		CIFAR100		SVHN	
	(1,2,7)	(0,5,5)	(1,2,7)	(0,5,5)	(1,2,7)	(0,5,5)	(1,2,7)	(0,5,5)
FedAvg-Lower (McMahan et al., 2017)	77.56 $\pm$ 0.17	84.89 $\pm$ 0.13	41.80 $\pm$ 0.48	53.05 $\pm$ 0.35	27.26 $\pm$ 0.06	19.40 $\pm$ 0.13	65.73 $\pm$ 0.28	70.81 $\pm$ 0.15
FedAvg + FixMatch (Sohn et al., 2020)	85.96 $\pm$ 0.39	86.76 $\pm$ 0.18	46.02 $\pm$ 0.16	74.56 $\pm$ 0.02	30.04 $\pm$ 0.09	25.14 $\pm$ 0.21	82.83 $\pm$ 0.92	85.35 $\pm$ 0.12
FedAvg + FreeMatch (Wang et al., 2023)	85.13 $\pm$ 2.45	86.30 $\pm$ 0.76	45.23 $\pm$ 3.02	73.29 $\pm$ 1.86	29.88 $\pm$ 0.58	24.59 $\pm$ 0.26	79.20 $\pm$ 3.71	84.64 $\pm$ 1.93
FedAvg + SoftMatch (Chen et al., 2023)	<u>87.73</u> $\pm$ 0.45	<u>89.85</u> $\pm$ 0.85	43.51 $\pm$ 0.87	75.12 $\pm$ 0.17	29.88* $\pm$ 0.52	20.13 $\pm$ 0.92	78.92 $\pm$ 0.87	85.24 $\pm$ 0.30
RSCFed (Liang et al., 2022)	81.20 $\pm$ 5.41	84.35 $\pm$ 1.83	40.20 $\pm$ 4.14	52.78 $\pm$ 1.45	24.12 $\pm$ 0.98	18.08 $\pm$ 1.05	63.37 $\pm$ 6.72	69.63 $\pm$ 2.97
CBAFed (Li et al., 2023)	86.73 $\pm$ 0.14	87.09 $\pm$ 0.07	52.88 $\pm$ 0.27	73.22 $\pm$ 0.16	28.92 $\pm$ 0.11	24.12 $\pm$ 0.06	81.32 $\pm$ 0.18	81.76 $\pm$ 0.27
Twin-Sight (Zhang et al., 2024)	86.95 $\pm$ 1.23	87.66 $\pm$ 0.48	54.84 $\pm$ 2.63	76.35 $\pm$ 0.87	30.79* $\pm$ 0.49	28.77 $\pm$ 0.32	83.15 $\pm$ 1.75	86.68 $\pm$ 0.86
FedGGp	88.59 $\pm$ 2.21	<u>89.02</u> $\pm$ 0.83	63.20 $\pm$ 3.24	79.90 $\pm$ 1.46	31.25* $\pm$ 0.81	31.99 $\pm$ 0.43	85.97 $\pm$ 2.67	<u>86.37</u> $\pm$ 2.02



(a) CIFAR10, (0,5,5)



(b) CIFAR100, (0,5,5)

Fig. 5. Convergence rate comparisons with (0,5,5) setting on CIFAR10 and CIFAR100. The compared methods are the best three methods for the corresponding scenario. In CIFAR10 experiments, all of those methods exhibit similar convergence rates, whereas on CIFAR100, higher-performing methods, such as Twin-Sight and FedGGp, converge more slowly and require additional communication rounds.

Table 2
Comparisons across different dirichlet factors on CIFAR10.

Method	Dir(0.1)	Dir(0.3)	Dir(0.8)	IID
FedAvg-Lower	43.65	53.05	56.14	58.45
FedAvg + FixMatch	55.60	74.56	79.46	83.33
FedAvg + SoftMatch	55.84	75.12	78.16	80.07
Twin-Sight	58.68	76.35	80.02	84.15
FedGGp	61.83	79.90	80.81	82.56

Table 3
Computational and communication overhead comparisons per round on CIFAR10.

Method	Communication cost	Computational complexity	Execution time
FedAvg-labeled	9.31 MB (1.0x)	28339.87 GFLOPs (1.0x)	1.4 min (1.0x)
FedAvg + FixMatch	9.31 MB (1.0x)	68020.25 GFLOPs (2.4x)	2.31 min (1.65x)
FedAvg + SoftMatch	9.31 MB (1.0x)	68020.25 GFLOPs (2.4x)	2.31 min (1.64x)
Twin-Sight	19.62 MB (2.21x)	161778.06 GFLOPs (5.71x)	2.63 min (1.88x)
FedGGp	9.31 MB (1.0x)	68020.26 GFLOPs (2.4x)	2.34 min (1.67x)

Table 4
Comparisons of accuracy and JS divergence across different rebalancing factors.

Factors	CIFAR10		CIFAR100	
	Accuracy	JS Divergence	Accuracy	JS Divergence
α_1	72.40	0.72	7.56	0.63
α_2	79.34	0.03	30.17	0.31
α_3	79.21	0.02	31.34	0.28

evident when compared to Twin-Sight, where the dual-model design necessitates a twofold increase in resource requirements.

4.3. Study of rebalancing factors

In this section, we discuss the impact of different rebalancing factors α_1 , α_2 , and α_3 on correcting prediction bias. The goal of rebalancing is to bring the prediction confidence as close as possible to a uniform distribution. To further evaluate the performance across different α , we introduce a second metric, using JS divergence to measure the alignment between the updated Φ and $P(U)$ at each round.

As shown in Table 4, despite α_2 and α_3 performing similarly on CIFAR10, the gap between them becomes more pronounced on CIFAR100. We attribute this to the fact that, as the number of classes increases, the stability of unidirectional expansion becomes lower compared to bidirectional scaling. Moreover, α_1 is prone to numerical instability due to the lack of a smoothing mechanism. The extremely high JS divergence observed in both the CIFAR10 and CIFAR100 experiments further supports this conclusion.

Furthermore, we analyzed the impact of the momentum update hyperparameter β on the experimental results. The experiments were evaluated on CIFAR10 and CIFAR100 under different values of $\beta \in \{0, 0.5, 0.9, 0.99\}$, as illustrated in the Fig. 6. It can be observed that the value of $\beta = 0.9$ yields superior performance in scenarios of greater complexity. As the complexity of the task decreases, an increase in the value of the parameter β becomes advantageous. We attribute this to the benefit of higher adaptive thresholds, which stems from the reduction in the boundary gap of Φ .

Table 5
Ablation studies with different pseudo-labeling components.

Dataset	Baseline		Ours		Accuracy
	Fixed T.	C.R.	Adaptive T.	Soft C.R.	
CIFAR10	✓	✓	-	-	74.53
	✓	-	-	✓	76.75
	-	✓	✓	-	78.65
	-	-	✓	✓	78.93
CIFAR100	✓	✓	-	-	26.88
	✓	-	-	✓	23.40
	-	✓	✓	-	30.86
	-	-	✓	✓	32.22

Table 6
Comparisons across different κ on CIFAR10 and CIFAR100.

Dataset	$\kappa = 1$	$\kappa = 2$	$\kappa = 3$	$\kappa = 5$	$\kappa = 10$
CIFAR10	78.65	79.98	79.78	78.89	78.56
CIFAR100	30.86	31.27	31.37	31.75	30.14

4.4. Ablation study

We have also conducted ablation studies to explore the impact of the two core components of the pseudo-labeling strategy: the Globally Guided Adaptive Threshold (Adaptive T.) and the Soft Consistency Regularization (Soft C.R.). The experiments also introduced two baseline configurations: Fixed Threshold (Fixed T.) and Consistency Regularization (C.R.). Among them, Adaptive T. and Fixed T. belong to the thresholding components, while C.R. and Soft C.R. represent the consistency regularization components. We combined these methods in ablation tests to demonstrate the superiority of the proposed approach.

The experimental results were presented in Table 5. In terms of thresholding strategies, Adaptive T. shows a super improvement over Fixed T. in both CIFAR10 and CIFAR100. For consistency regularization, Soft C.R., which incorporates a recovery mechanism, does not demonstrate a significant advantage over C.R. when used independently. This suggests that relaxing consistency constraints can yield dual effects, necessitating a well-designed thresholding strategy to ensure effectiveness. Notably, Adaptive T. provides a foundation for pseudo-labeling recovery, as evidenced by FedGGp, which combines both components and delivers the most robust overall performance.

Another notable observation is that applying Soft C.R. in conjunction with Adaptive T. does not yield substantial improvement in the CIFAR10 scenario. The reason is that the number of tail classes in the globally guided class division is extremely limited. To substantiate this assertion, we conducted experiments on the Top-k mechanism with the hyperparameter κ taking on values in $\{1, 2, 3, 5, 10\}$, as shown in Table 6. The results demonstrate that model accuracy on CIFAR10 remains stable despite changes in κ . However, in CIFAR100, an excessive decline in performance is observed when κ is enlarged. This can be attributed to the incorporation of low-confidence pseudo-labels in Soft C.R., which impedes the learning of the corresponding classes.

5. Conclusion

In this work, we propose FedGGp, a novel FSSL framework designed to address the challenges posed by inherently biased local predictions. FedGGp constructs globally calibrated probabilities to decouple the adjustment of local pseudo-labeling strategy from biased local predictions, providing probability space guidance free of prior bias for clients. Compared to the frequency-based guidance, this approach is more equitable and less susceptible to interference from imbalanced labeled distributions. Furthermore, FedGGp constructs stratified pseudo-labelings based on category predictions, which achieves class-wise balanced compensation through differentiated adaptive thresholds and regularizations. The

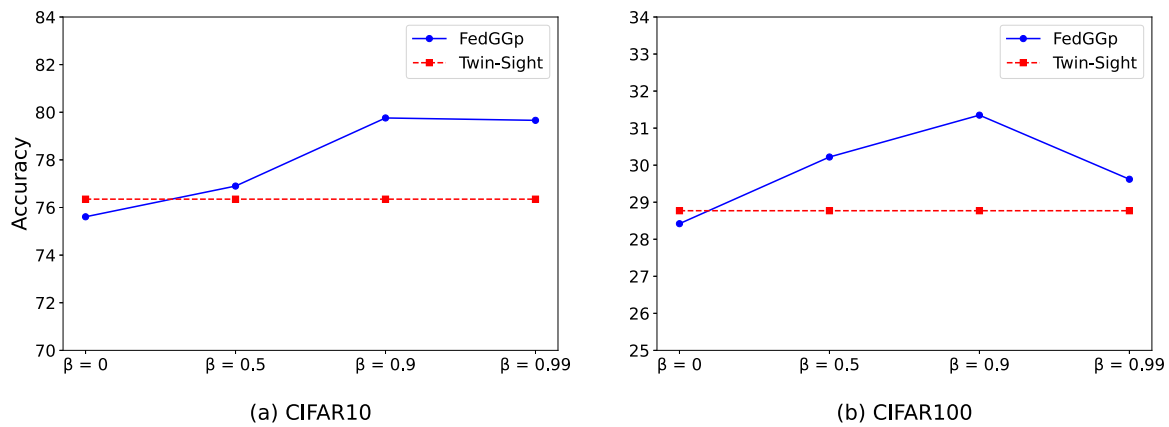


Fig. 6. Comparison across different β on CIFAR10 and CIFAR100, with the suboptimal method Twin-sight as a baseline.

proposed soft consistency regularization ensures the retention of training contributions from filtered-out samples near decision boundaries, which in turn helps to effectively mitigate extreme class shifts. This also suggests that consistency-filtered samples, which have previously been discarded, can actually exert a positive influence on the learning of underrepresented classes in FSSL. Experimental results across diverse datasets demonstrate the superior performance of FedGGp compared to existing state-of-the-art methods. Moreover, the flexible architecture of FedGGp facilitates seamless integration with other FL techniques, making it broadly applicable to various learning scenarios. In future work, we plan to focus on reducing the instability of pseudo-labeling and enhancing fairness for rare classes.

CRedit authorship contribution statement

Yuan Xi: Methodology, Conceptualization, Writing – original draft; **Qiong Li:** Project administration, Supervision; **Haokun Mao:** Writing – review & editing.

Data availability

We have used publicly available datasets. The code of FedGGp can be accessed at <https://github.com/whatement/FedGGp.git>.

Declaration of competing interest

The authors declare that they have no competing interests regarding the content of this manuscript. There are no financial or personal relationships with individuals or organizations that could inappropriately influence the work presented in this study.

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Supplementary material

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